SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

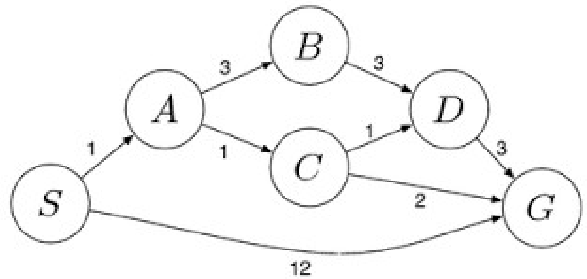
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

   The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

1st)Start with both jugs empty:

1 **(0,0)** – Fill the 3-gallon jug → **(0,3)**
2 Pour the 3-gallon jug into the 4-gallon jug → **(3,0)**
3 Fill the 3-gallon jug again → **(3,3)**
4 Pour from the 3-gallon jug into the 4-gallon jug until the 4-gallon jug is full. The 4-gallon jug needs only 1 more gallon, so 2 gallons remain in the 3-gallon jug → **(4,2)**
5 Empty the 4-gallon jug onto the ground → **(0,2)**
6 P0ur the 2 gallons from the 3-gallon jug into the 4-gallon jug → **(2,0)**

2nd)Percepts are the information the rover receives through its sensors.

1. High-resolution camera images
2. Stereo and navigation camera data
3. Temperature, pressure, and weather measurements
4. Soil and rock composition from spectrometers
5. Chemical and mineral analysis data
6. Terrain and obstacle information (rocks, slopes, sand)
7. Wheel position, speed, and motor status
8. Battery level and power status
9. Position and orientation (using onboard navigation)
10. Commands received from mission control on Earth

The Mars Rover operates in the following environment:

- **Partially observable:** The rover cannot sense everything about its surroundings at once.
- **Single-agent:** It performs its mission independently.
- **Sequential:** Current actions affect future situations.
- **Dynamic:** Environmental conditions such as dust storms, lighting, and terrain can change.
- **Continuous:** Movement, time, and sensor readings are continuous.
- **Unknown:** The rover encounters new terrain and unexpected obstacles that were not known beforehand.

The Mars Rover can:

1. Move forward, backward, and turn
2. Navigate around obstacles
3. Capture images and videos

4. Drill into rocks and collect soil samples
5. Analyze samples using onboard scientific instruments
6. Measure atmospheric conditions
7. Store collected samples

The rover's performance can be evaluated by:

1. Successfully reaching exploration targets
2. Accuracy of navigation and obstacle avoidance
3. Number and quality of scientific samples collected
4. Accuracy of scientific measurements and analyses
5. Amount of useful data transmitted to Earth
6. Efficient use of power and resources
7. Ability to operate for a long mission duration
8. Reliability in handling unexpected situations and faults

A **utility-based, model-based agent architecture** is the most suitable.
Reason:

- The rover maintains an internal model of its environment because it cannot observe everything directly.
- It evaluates multiple possible actions and selects the one that maximizes mission objectives, such as scientific value, safety, and energy efficiency.
- It can adapt to changing terrain, avoid hazards, and make autonomous decisions when communication with Earth is delayed (typically 4–24 minutes one way).
- It balances competing goals, such as collecting valuable samples while conserving power and minimizing risk.
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Conclusion:
A **model-based utility agent** provides the best combination of autonomy, planning, environmental awareness, and decision-making required for successful exploration and scientific investigation on Mars.

1. Problem Formulation

a) Initial State

- An empty 8 × 8 chessboard with no queens placed.

b) State Space

- A state is any arrangement of 0 to 8 queens on the board.
- A common representation is an 8-element array where the index represents the **column** and the value represents the **row** of the queen.

- Example: **[1, 5, 8, 6, 3, 7, 2, 4]**

c) Actions (Operators)

- Place a queen in the next column on any row that does not violate the constraints.
- During search, move or reposition queens if using local search algorithms.

d) Goal Test

A solution is reached when:

- All 8 queens are placed.
- No two queens share:
  - the same row,
  - the same column,
  - the same diagonal.

e) Path Cost

- Usually each queen placement has a cost of **1**.
- Since every solution requires placing 8 queens, all solutions have equal path cost.

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2. Search Space

- Without constraints:
  - Each of the 8 columns can have 8 possible row positions.
  - Total states = $\mathbf{8^8 = 16{,}777{,}216}$.
- By restricting to one queen per column:
  - Total arrangements become $\mathbf{8! = 40{,}320}$.
- Search algorithms further reduce the search space by pruning invalid states.

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3. Search Strategies

a) Breadth-First Search (BFS)

- Explores all states level by level.
- **Advantages:** Complete; guarantees finding a solution.
- **Disadvantages:** Very high memory usage and inefficient for large search spaces.

b) Depth-First Search (DFS)

- Places queens column by column until a conflict occurs.
- Backtracks when necessary.
- **Advantages:** Low memory usage.
- **Disadvantages:** May explore many dead ends before finding a solution.

c) Backtracking Search (Most Common)

- A specialized form of DFS.
- Places one queen at a time.
- Immediately rejects placements that violate constraints.
- **Advantages:** Efficient and widely used for the 8-Queens problem.

d) Hill Climbing (Local Search)

- Starts with a complete board configuration.
- Moves queens to reduce the number of conflicts.
- **Advantages:** Fast in many cases.
- **Disadvantages:** Can become stuck in local optima.

e) Simulated Annealing / Genetic Algorithm

- Useful for larger N-Queens problems.
- Escapes local optima by allowing occasional non-improving moves or evolving candidate solutions.

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4. Constraints

A valid configuration must satisfy:

- Exactly one queen in each row.
- Exactly one queen in each column.
- No two queens on the same diagonal.

Diagonal conflict condition:

- Two queens at positions (r_1, c_1) and (r_2, c_2) conflict if:

$|r_1 - r_2| = |c_1 - c_2|$

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5. Example of a Non-Solution

Suppose the queens are arranged so that:

- The queen in the **first column** and the queen in the **last column** lie on the same diagonal.

Although there are 8 queens on the board, this **is not a valid solution** because two queens can attack each other diagonally.

Summary

Problem Component	Description
Initial State	Empty 8 × 8 chessboard
State	Placement of queens on the board
Actions	Place or move a queen while satisfying constraints
Goal Test	Eight queens placed with no attacks
Path Cost	One cost per queen placement (uniform)
State Space	Up to 8^8 states (reduced to 8! with one queen per column)

4rt)Problem-solving agent: Goal-Based Agent

Reason:

A goal-based agent selects the best cab (Mini, Micro, Sedan, Shared, Prime, etc.) to achieve the goal of reaching the destination based on the customer's preference.

Pseudocode:

Start
Input source and destination
Display available cab types
Customer selects a cab type
Book the selected cab
Confirm booking
End

5th)Algorithm: Uniform Cost Search (UCS)

Uniform Cost Search (UCS) finds the **least-cost path** from the start warehouse **S** to the destination **G** by always expanding the node with the **lowest cumulative cost** first. It guarantees the optimal path when all route costs are positive.

Pseudocode:

```
Start
Add S to priority queue (cost = 0)
While queue is not empty
   Remove node with lowest cost
   If node = G, return path
   Add neighboring nodes with updated costs
End
```